

A Proactive Planning Framework for Valuing Active Distribution System Investments Under Uncertainty

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NOMENCLATURE

Indices and sets

$a, b \in \mathcal{A}$	Planning alternatives
$c \in \mathcal{C}$	Decision criteria
$\mathcal{D}_a$	Decision-stability region, alt. a
$e \in \mathcal{E}$	Economic indicators
$f \in \mathcal{F}$	Decision-making functionals
$k \in \mathcal{K}$	Technology building blocks
$\mathbb{N}, \mathbb{R}$	Natural and real numbers
n	Sampled weight vector index, $n \leq N_\Omega$
$s \in \mathcal{S}$	Planning scenarios
$t \in \mathcal{T}_y$	Operational periods, planning year y
$u \in \mathcal{U}$	Uncertain variables
$\mathcal{W}$	Scenario-weight space
$y \in \mathcal{Y}$	Planning years
Ξ_s	Set of uncertainty realisations, scenario s

Scenario weights

Ω	Vector of scenario weights, $\Omega \in \mathcal{W}$ [p.u.]
$\hat{\Omega}$	Baseline (initial) scenario weight vector [p.u.]
$\Omega^{(n)}$	n -th sampled scenario weight vector [p.u.]
ω_s	Scenario weight, scenario s [p.u.]
ω_x, ω_y	Weights of the scenarios on the x - and y -axes of a stability-region panel [p.u.]

Economic indicators, matrices and vectors

$\mathbf{EI}^e$	Indicator matrix e , $\mathbb{R}^{ \mathcal{A} \times \mathcal{S} }$ [\$]
$\mathbf{ei}_a^e$	Row a of $\mathbf{EI}^e$, $\mathbb{R}^{ \mathcal{S} }$ [\$]
$\mathbf{NPC}_{a,s}$	Net present total system cost, alt. a , sce. s [\$]
$\mathbf{NPC}_{:,s}$	Column s of $\mathbf{NPC}$ (all alternatives, sce. s) [\$]
$\mathbf{R}_{a,s}(\cdot)$	Regret of an indicator, alt. a , sce. s , e.g., $\mathbf{R}(\mathbf{NPC})$ [\$]
$\mathbf{USE}_{a,s}$	Unserved energy cost, alt. a , sce. s [\$]
$\mathbf{WR}_{a,s}(\cdot)$	Weighted regret of an indicator, alt. a , sce. s (depends on Ω), e.g., $\mathbf{WR}(\mathbf{NPC}, \Omega)$ [\$]

Parameters

M	Number of short-term increments
N	Number of long-term increments
N_Ω	Number of sampled scenario weight vectors
n_a	Number of blocks in alt. a
y_0	Initial planning year [yr]
α	Risk level of CVaR^α [%]
ΔY	Long-term increment [yr]
Δy	Short-term increment [yr]

Variables and functions

$a^*(\Omega)$	Recommended alt. under weights Ω
$\mathbf{C}_{a,s,y}^{\text{inv}}$	Total investment cost, alt. a , sce. s , yr. y [\$/yr]
$\mathbf{C}_{a,s,y}^{\text{op}}$	Total operating cost, alt. a , sce. s , yr. y [\$/yr]
$c^{\text{op}}(\cdot)$	Operating cost function, period t [\$]
$\mathbf{CC}_{a,s,y}$	Customer DER curtailment cost, alt. a , sce. s , yr. y [\$/yr]
CVaR_a^α	Expected shortfall at risk level α , alt. a [\$]
$d(\cdot)$	Nodal power balance constraints
$\mathbb{E}_a$	Scenario-weighted mean, alt. a [\$]
$\hat{F}_a^{f,e}(v)$	Empirical CDF, funct. f , ind. e , alt. a [p.u.]
$g(\cdot)$	Network and DER operating limits
$h(\cdot)$	Inter-temporal coupling constraints
$\max_s$	Worst-case outcome across scenarios, alt. a [\$]
$\mathbf{USE}_{a,s,y}$	Unserved energy cost, alt. a , sce. s , yr. y [\$/yr]
v	Functional threshold value [\$]
$\mathbf{x}_t$	Operational decisions, period t
$\gamma_{k,s,y}$	Annualised cost, block k , sce. s , yr. y [\$/yr]
$\gamma_{k,s,y}^{\text{fix}}$	Fixed capital cost, block k , sce. s , yr. y [\$/yr]
$\gamma_{k,s,y}^{\text{var}}$	Usage cost, block k , sce. s , yr. y [\$/yr]
$\xi_{s,y,u}$	Realisation of uncertain var. u , sce. s , yr. y
$\xi_{s,y,t}$	Vector of operating conditions, sce. s , yr. y , period t
$\pi_a^{f,e}$	Confidence in alt. a (share of samples in which it is recommended), funct. f , ind. e [p.u.]
$\rho_{k,s,y}$	Utilisation, block k , sce. s , yr. y [p.u.]
$\phi(y, s)$	System pathway assigned to yr. y , sce. s
$\phi_u(y, s)$	Value of uncertain var. u on pathway $\phi(y, s)$
$\Psi^f(\cdot)$	Decision-making functional [\$]
$\psi_a^f(\cdot)$	Functional value, alt. a [\$]
$\mathbf{1}[\cdot]$	Indicator function [0/1]

Case-study building blocks

DR_1	DR contract, first price band (case study 1)
DR_2	DR contract, first two price bands (case study 1)
DR_3	DR contract, all three price bands (case study 1)
$\text{DS}^{\text{L/M/H}}$	Distributed storage at low/medium/high level (case study 2)
ESS_1	Two-hour battery ESS (case study 1)
ESS_2	Four-hour battery ESS (case study 1)

Abbreviations

A\$	Australian dollars
ADS	Active distribution system
AEMO	Australian Energy Market Operator
CBTS	Cranbourne Terminal Station
CDF	Cumulative distribution function
CVaR	Conditional value at risk (expected shortfall)
DER	Distributed energy resources
DHW	Domestic hot water

DMF	Decision-making functional
DN	Do nothing
DR	Demand response
DS	Distributed storage (low/medium/high)
ESS	Energy storage system
EU	European Union
EV	Electric vehicle
ISP	Integrated System Plan
MCA	Multi-criteria analysis
NNS	Non-network solutions
NPC	Net present total system cost
Nwk	Network (building block)
OPF	Optimal power flow
PV	Photovoltaic
RO	Real options
UK	United Kingdom
US	United States
USE	Unserved energy (cost)
WR	Weighted regret
\$	Generic currency unit (case-study values in A\$)